Standard Operating Procedure for Resolving Critical Hardware/Environmental Faults in Cisco PE MAR-PE-13 at Marathahalli Core

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_4636

Description

The PE role in Cisco's MAR-PE-13 node at Marathahalli Core is susceptible to hardware/environmental faults due to the inherent complexities of its ASIC memory parity errors, line card crashes, and power supply redundancy loss mechanisms. This is largely attributed to the Advanced Services Interface (ASI) module's capacity for processing high volumes of data at extreme speeds, which in turn necessitates an unyielding reliability in its hardware components to maintain the integrity of the data plane. Moreover, the PE node's dependence on line cards for routing and forwarding decisions further exacerbates the likelihood of hardware failures. Additionally, the redundancy in power supplies, although designed to mitigate such failures, can itself become a single point of failure if not properly configured or maintained. Consequently, it is crucial to promptly identify and rectify any hardware/environmental faults in the PE role to prevent network disruptions and ensure the overall reliability of the Marathahalli Core.

Pre-requisite Checks

- Dispatch remote hands to Marathahalli Core to ensure physical accessibility to the node.
- Verify the status of redundant links to ensure optimal network redundancy.
- Check the power supply redundancy to confirm its operational status.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

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Command: show diag, Expected Output: Diagnosis information for the node, including any hardware errors or malfunctions., Description: This command isolates the control plane's diagnostic capabilities to identify any potential hardware issues within the node. The output will provide detailed information regarding the node's hardware configuration, including any errors or malfunctions that might be contributing to the fault.
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Command: show tech-support, Expected Output: Detailed technical support information for the node, including hardware and software configurations., Description: This command activates the node's technical support capabilities to gather detailed information regarding the node's hardware and software configurations. The output will provide a comprehensive overview of the node's technical specifications, including any potential issues that might be contributing to the fault.

Command: show power redundancy, Expected Output: Power supply redundancy status information for the node., Description: This command isolates the node's power supply redundancy status to confirm its operational status. The output will provide detailed information regarding the power supply redundancy, including any issues that might be contributing to the fault.
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Resolution Steps

Step 1: Step: Replace faulty hardware components, Command: replace slot , Description: This step involves replacing any faulty hardware components identified during the diagnostic process. The command 'replace slot ' is used to initiate the replacement process, where corresponds to the specific slot containing the faulty hardware component.

Step 2: Step: Reconfigure power supply redundancy, Command: configure power redundancy, Description: This step involves reconfiguring the power supply redundancy to ensure its optimal operation. The command 'configure power redundancy' is used to initiate the reconfiguration process, which will ensure that the power supply redundancy is properly configured to prevent any future faults.

Step 3: Step: Verify hardware configuration, Command: show diag, Description: This step involves verifying the hardware configuration to ensure that it is properly configured to prevent any future faults. The command 'show diag' is used to isolate the control plane's diagnostic capabilities to identify any potential hardware issues within the node.

Post-Check Verification

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Command: show power redundancy, Expected Output: Power supply redundancy status information for the node, confirming its optimal operation., Description: This command isolates the node's power supply redundancy status to confirm its optimal operation. The output will provide detailed information regarding the power supply redundancy, including its status and any potential issues that might be contributing to the fault.

Command: show diag, Expected Output: Diagnosis information for the node, confirming its optimal hardware configuration., Description: This command isolates the control plane's diagnostic capabilities to identify any potential hardware issues within the node. The output will provide detailed information regarding the node's hardware configuration, including any errors or malfunctions that might be contributing to the fault.
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Roll-back Procedures

- Step: Restore original hardware configuration, Command: configure replace , Description: This step involves restoring the original hardware configuration to its previous state. The command 'configure replace ' is used to initiate the restoration process, where corresponds to the specific slot containing the replaced hardware component.

- Step: Revert power supply redundancy configuration, Command: configure power redundancy revert, Description: This step involves reverting the power supply redundancy configuration to its original state. The command 'configure power redundancy revert' is used to initiate the reversion process, which will ensure that the power supply redundancy is properly configured to prevent any future faults.